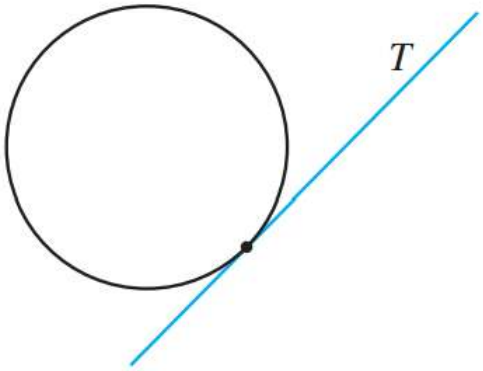
Ch2. Derivatives

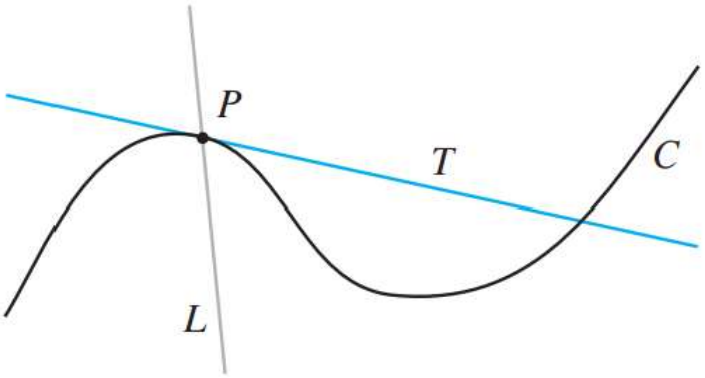
§2.1 Derivatives and Rates of Change

[The Tangent Problem]

Introduction " tangent " <- tangens (Latin word) == " touching "

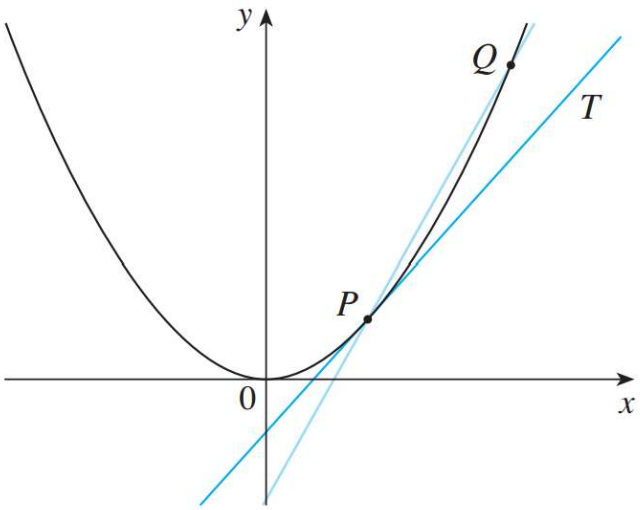


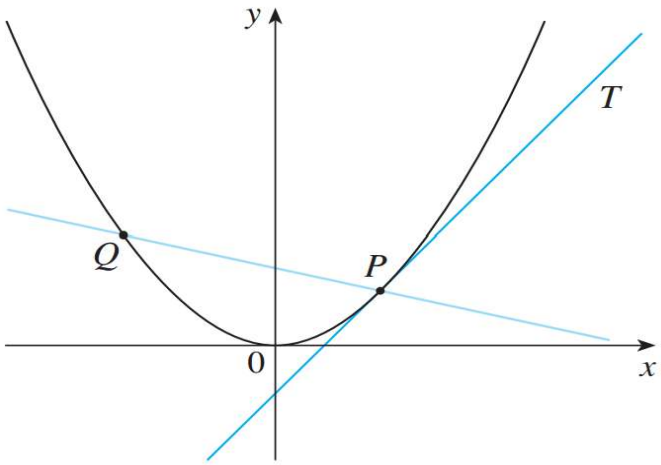
(a)



(b)

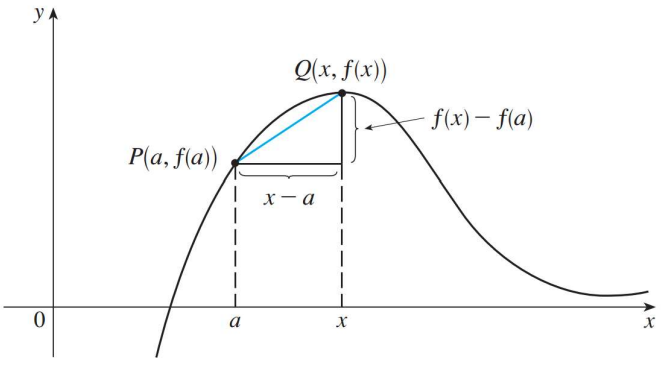
Example Find an equation of the tangent line to the parabola $y = x^2$ at the point $P(1,1)$.

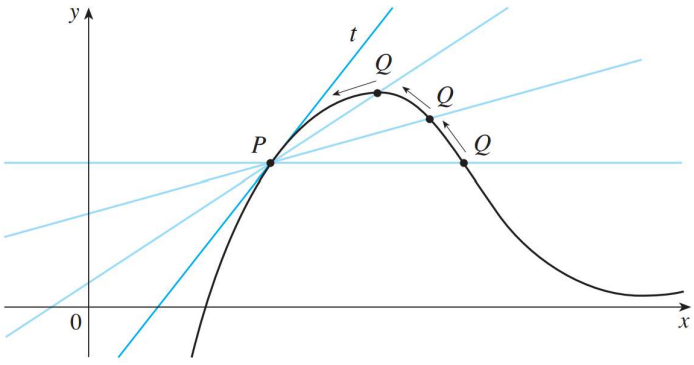




Ch2. Derivatives

Let a curve $C : y = f(x)$ and $P(a, f(a))$ be a point on the curve.



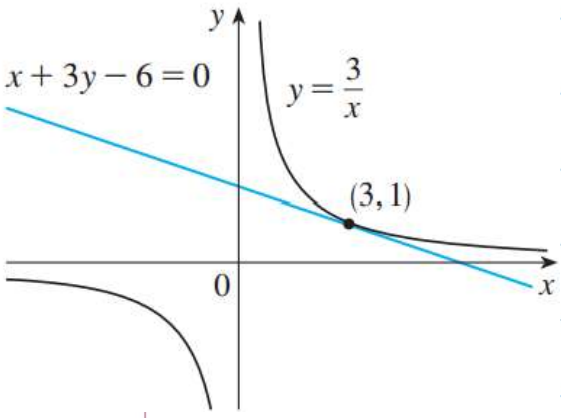


Definition The **tangent line** to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with slope

$$m = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

provided that this limit exists.

Example Find an equation of the tangent line to the hyperbola $y = \dfrac{3}{x}$ at the point $P(3, 1)$.



Ch2. Derivatives

[Derivatives]

Definition The **derivative of a function** f **at a number** a**,** denoted by $f'(a)$, is

$$f'(a)=\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}=\lim_{x\to a}\frac{f(x)-f(a)}{x-a}$$

if the limit exists.

Example Find the derivative of the function $f(x)=x^2-8x+9$ at the number a.

Definition (revisited)
The tangent line to $y=f(x)$ at $(a,f(a))$ is the line through $(a,f(a))$ whose slope is equal to $f'(a)$, the derivative of f at a.

Example Find an equation of the tangent line to the parabola $y=x^2-8x+9$ at the point $(3,-6)$.

Note The derivative $f'(a)$ is the instantaneous rate of change of $y=f(x)$ with respect to x when $x=a$.

Ch2. Derivatives

§2.2 The derivative as a function

recall $f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$.

Definition the **derivative of** f : $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

Example If $f(x) = x^3 - x$, find a formula for $f'(x)$.

Example If $f(x) = \sqrt{x}$, find a formula for $f'(x)$ and its domain.

Ch2. Derivatives

Example If $f(x)=\dfrac{1-x}{2+x}$, find a formula for $f'(x)$.

[Notations]

Notation $$f'(x)=y'=\frac{dy}{dx}=\frac{df}{dx}=\frac{d}{dx}f(x)=Df(x)=D_xf(x)$$

$D=\dfrac{d}{dx}$: **differentiation operators**

$\dfrac{d}{dx}\square$ means "differentiate $\square$ with respect to x"

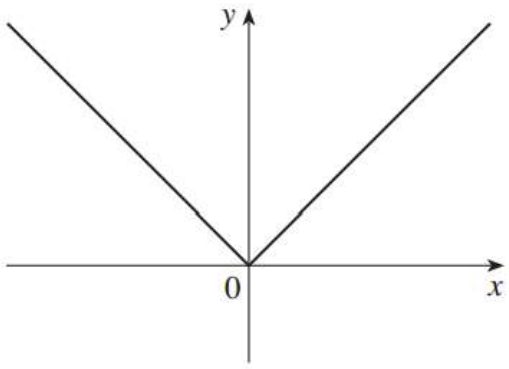
Note $$\frac{dy}{dx}=\lim_{\triangle x\to 0}\frac{\triangle y}{\triangle x}$$

Notation $\left.\dfrac{dy}{dx}\right|_{x=a}$, $\left.\dfrac{dy}{dx}\right]_{x=a}$: $f'(a)$

Ch2. Derivatives

Definition A function f is **differentiable at** a if $f'(a)$ exists. It is **differentiable on an open interval** (a,b)[or $(a,\infty),(-\infty,a),(-\infty,\infty)$] if it is differentiable at every number in the interval.

Example Where is the function $f(x)=|x|$ differentiable?

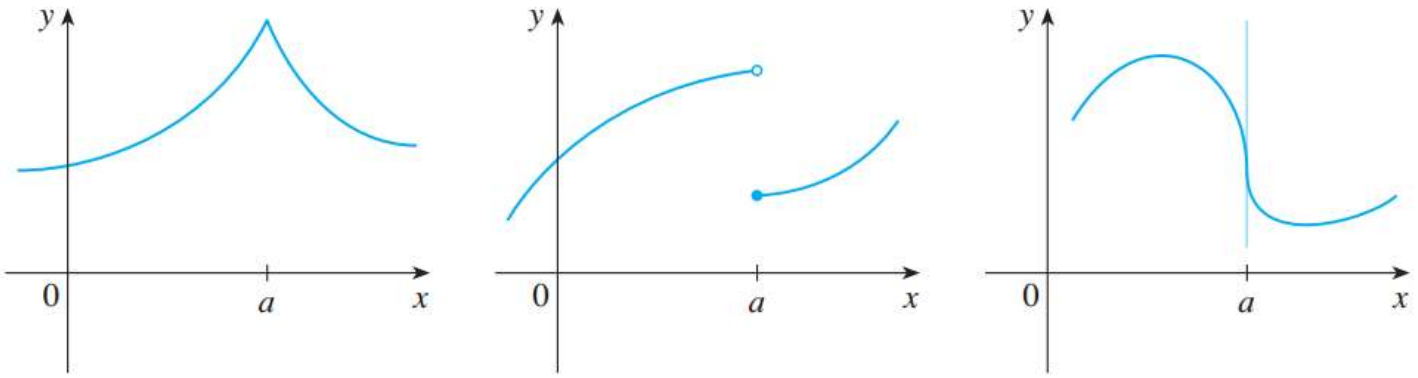


(a) $y=f(x)=|x|$

Theorem If f is differentiable at a, then f is continuous at a.

[How can a function fail to be differentiable?]

Three ways for f not to be differentiable at a :



(a) A corner (b) A discontinuity (c) A vertical tangent

[Higher Derivatives]

Note If f is a differentiable function, then its derivative f' is also a function.

example : $f(x)=x^3+3x^2$.

the derivative of a function $f'(x)$, denoted by $f''(x)$, is called the **second derivative** of f

Notation $f''(x)=\dfrac{d}{dx}\left(\dfrac{dy}{dx}\right)=\dfrac{d^2y}{dx^2}$

$f'''(x)=\dfrac{d}{dx}\left(\dfrac{d^2y}{dx^2}\right)=\dfrac{d^3y}{dx^3}$: the **third derivative**

$y^{(n)}=f^{(n)}(x)=\dfrac{d}{dx}\left(\dfrac{d^{n-1}y}{dx^n}\right)=\dfrac{d^ny}{dx^n}$: the **nth derivative**

Example If $f(x)=x^3-x$, find $f''(x)$.

§2.3 Basic Differentiation Formulas

DERIVATIVE OF A CONSTANT FUNCTION

$$\frac{d}{dx}(c) = 0$$

$$\frac{d}{dx}(x) = 1$$

THE POWER RULE If n is a positive integer, then

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

THE POWER RULE (GENERAL VERSION) If n is any real number, then

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Example Evaluate the 1st derivatives of the following functions.

(a) $f(x) = x^6$

(b) $y = x^{2025}$

(c) $y = t^4$

(d) r^3

(e) $g(x) = \dfrac{1}{x}$

(f) $y = \sqrt[3]{x^2}$

(g) $h(x) = x\sqrt{x}$

[Linear combination of old functions]

THE CONSTANT MULTIPLE RULE If c is a constant and f is a differentiable function, then

$$\frac{d}{dx}[cf(x)] = c\frac{d}{dx}f(x)$$

THE SUM RULE If f and g are both differentiable, then

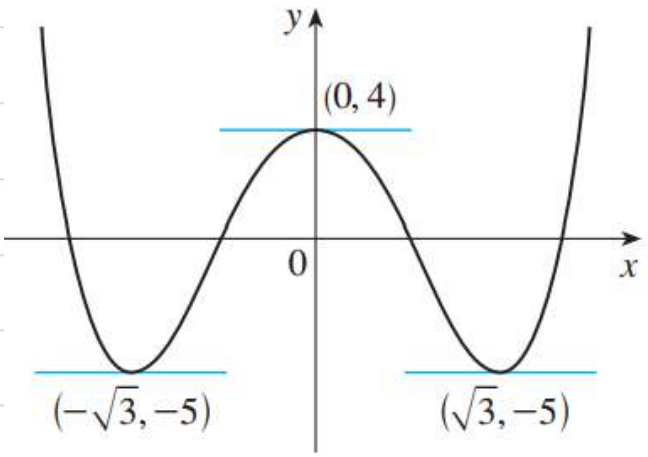
$$\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}f(x) + \frac{d}{dx}g(x)$$

THE DIFFERENCE RULE If f and g are both differentiable, then

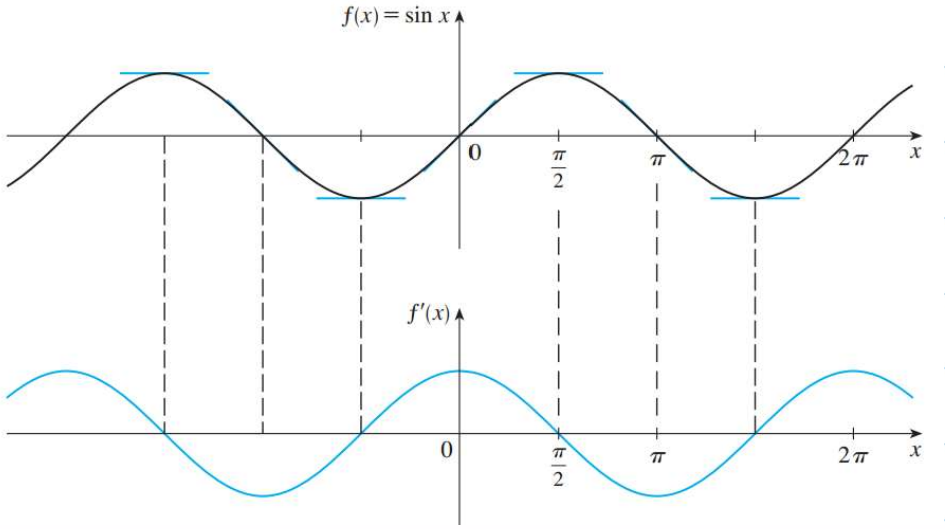
$$\frac{d}{dx}[f(x) - g(x)] = \frac{d}{dx}f(x) - \frac{d}{dx}g(x)$$

Example Compute y' if $y = x^8 + 12x^5 - 4x^4 + 10x^3 - 6x + 5$.

Example Find the points on the curve $y = x^4 - 6x^2 + 4$ where the tangent line is horizontal.


y
(0, 4)
0
x
$(-\sqrt{3}, -5)$
$(\sqrt{3}, -5)$

[The sine and cosine functions]



Note $\lim_{\theta \to 0} \frac{\sin\theta}{\theta} = 1$ $\lim_{\theta \to 0} \frac{\cos\theta - 1}{\theta} = 0$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

Example Differentiate $y = 3\sin\theta + 4\cos\theta$

Example Find the 27th derivative of $\cos x$.

§2.4 The Product and Quotient Rules

THE PRODUCT RULE If f and g are both differentiable, then

$$\frac{d}{dx}[f(x)g(x)] = f(x)\frac{d}{dx}[g(x)] + g(x)\frac{d}{dx}[f(x)]$$

Example Differentiate the following functions.

(a) $y = x^2 \sin x$

(b) $f(t) = \sqrt{t}\,(a + bt)$

Example If $h(x) = xg(x)$ and it is known that $g(3) = 5$ and $g'(3) = 2$, find $h'(3)$.

THE QUOTIENT RULE If f and g are differentiable, then

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)\frac{d}{dx}[f(x)] - f(x)\frac{d}{dx}[g(x)]}{[g(x)]^2}$$

$$(fg)' = f'g + fg' \qquad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

Example Find y' if $y = \dfrac{x^2+x-2}{x^3+6}$.

Example Find an equation of the tangent line to the curve $y = \sqrt{x}/(1+x^2)$ at the point $(1,1/2)$.

DERIVATIVES OF TRIGONOMETRIC FUNCTIONS

$$\frac{d}{dx}(\sin x) = \cos x \qquad \frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\cos x) = -\sin x \qquad \frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x \qquad \frac{d}{dx}(\cot x) = -\csc^2 x$$

$y = \tan x = \dfrac{\sin x}{\cos x}$

Example Differentiate $f(x) = \dfrac{\sec x}{1+\tan x}$. For what values of x does the graph of f have a horizontal tangent.

§2.5 The Chain Rule

THE CHAIN RULE If f and g are both differentiable and $F = f \circ g$ is the composite function defined by $F(x) = f(g(x))$, then F is differentiable and F' is given by the product

$$F'(x) = f'(g(x)) \cdot g'(x)$$

In Leibniz notation, if $y = f(u)$ and $u = g(x)$ are both differentiable functions, then

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

Example Find $F'(x) = \sqrt{x^2+1}$.

Example Differentiate (a) $y=\sin(x^2)$ and (b) $y=(\sin x)^2=\sin^2 x$.

Example Differentiate

(a) $y=(x^3-1)^{100}$

(b) $y=\dfrac{1}{\sqrt[3]{x^2+x+1}}$

(c) $y=\left(\dfrac{t-2}{2t+1}\right)^9$

(d) $y=(2x+1)^5(x^2-x+1)^4$

(e) $y=\sin(\cos(\tan x))$

(f) $y=\sqrt{\sec x^3}$

§2.6 Implicit Differentiation

Definition

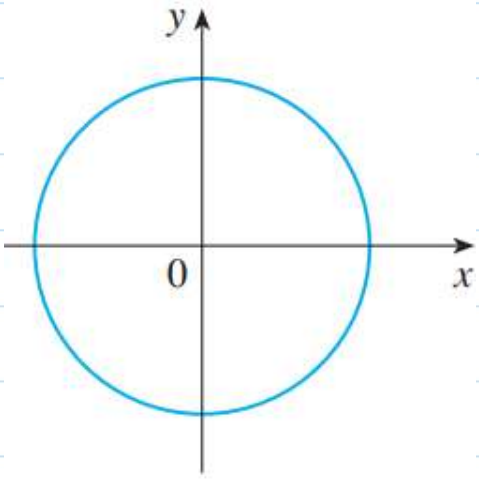
(a) A function f is an **explicit** function if it can be described by expressing one (dependent) variable explicitly in terms of another (independent) variable.

$$y = f(x)$$

(b) A function f is an **implicit** function if it cannot be described as one (dependent) variable in terms of another (independent) variable.

$$f(x,y) = 0 \text{ or } f(x,y) = c \ (c : \text{a constant})$$

(ex) $y = \sqrt{x^3+1}$ $\quad x^2+y^2 = 16$ $\quad x^3+y^3 = 6xy$



(a) $x^2+y^2=25$

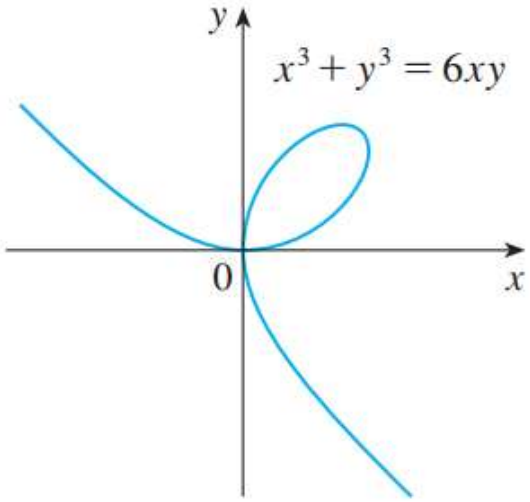


FIGURE 2 The folium of Descartes

Example

(a) Find $\dfrac{dy}{dx}$ if $x^2+y^2=25$.

(b) Find an equation of the tangent to the circle $x^2+y^2=25$ at the point $(3,4)$.

Example (a) Find y' if $x^3+y^3=6xy$.

(b) Find the tangent to the folium of Decartes $x^3+y^3=6xy$ at the point $(3,3)$.

(c) At what point in the first quadrant is the tangent line horizontal?

Example Find y' if $\sin(x+y)=y^2\cos x$.

Example Find y'' if $x^4+y^4=16$.

§2.7 Related Rates

Example Air is being pumped into a spherical balloon so that its volume increases at a rate of 100 cm^3/s. How fast is the radius of the balloon increasing when the diameter is 50cm?

Example A water tank has the shape of an inverted circular cone with base radius $2m$ and height $4m$. If water is being pumped into the tank at a rate of $2m^3/\min$, find the rate at which the water level is rising when the water is $3m$ deep.

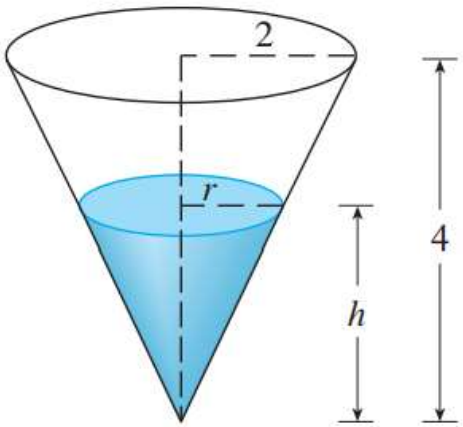


FIGURE 3